

Nanotechnology in cosmetics

DBS BIOTECHNOLOGY

5TH SEMESTER

03

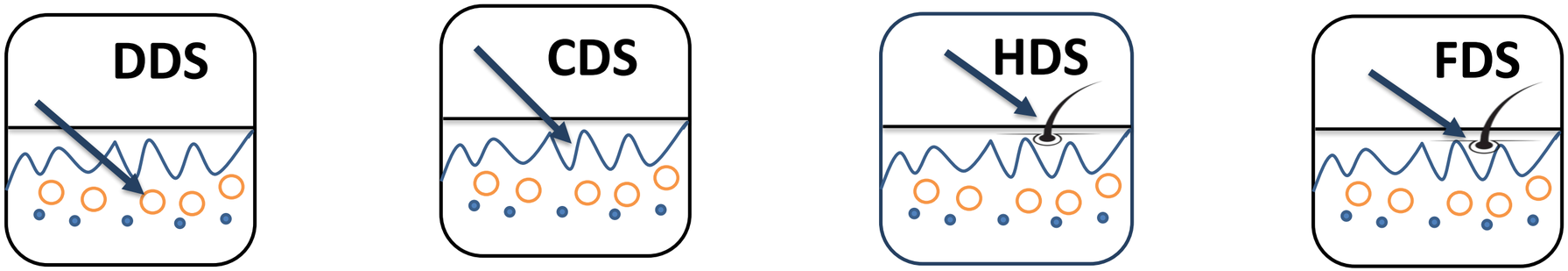
ESHA ZAMAN

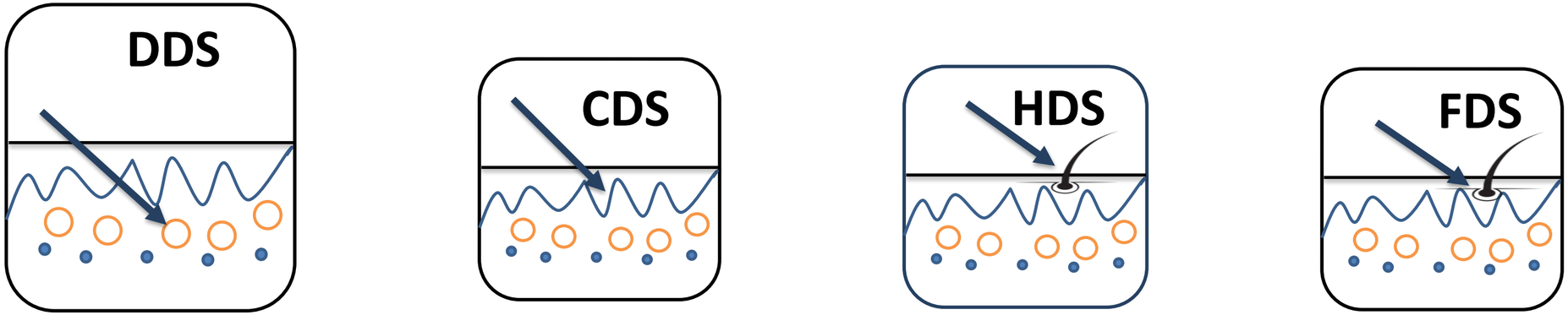
Delivery system for each cosmetic target

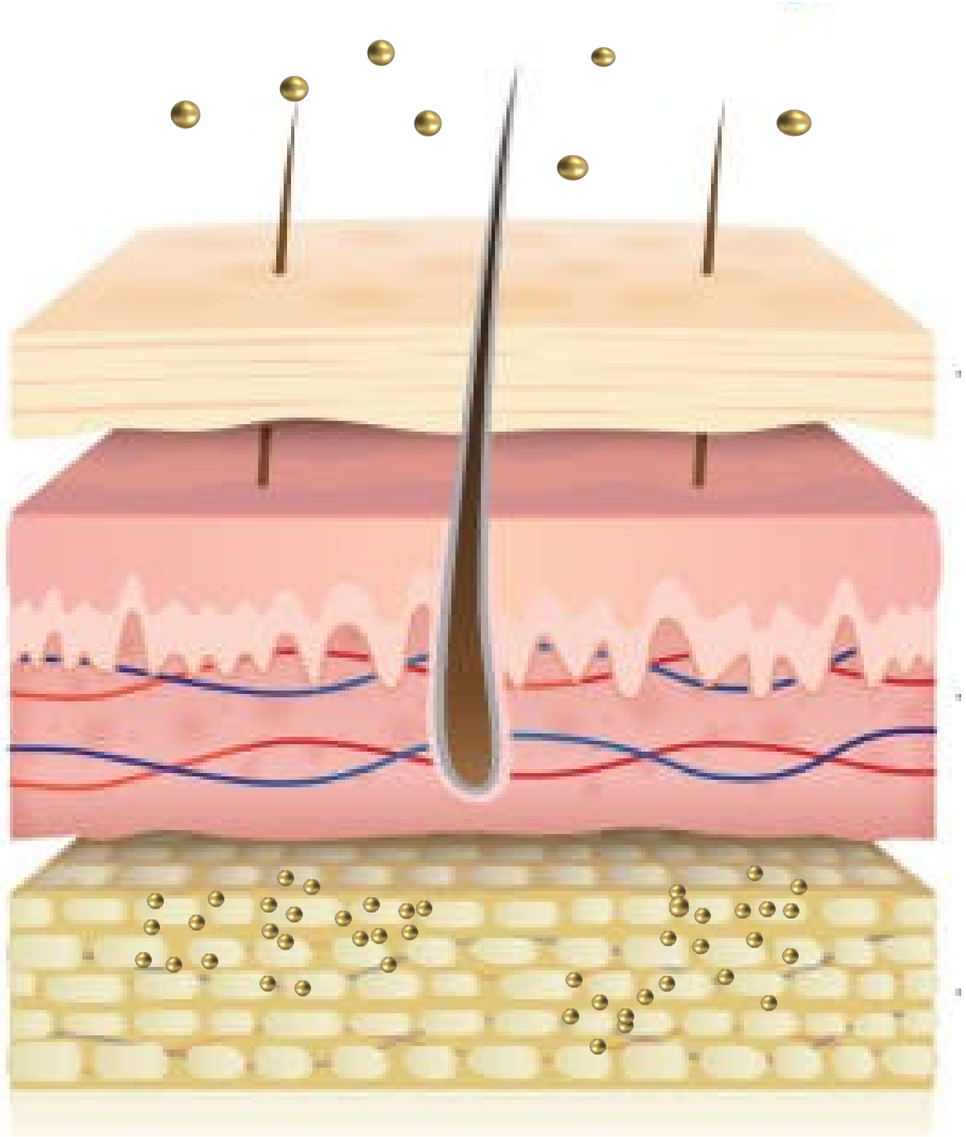
Nano based Sunscreens

Nano based topical acne treatment

Delivery system for each cosmetic target

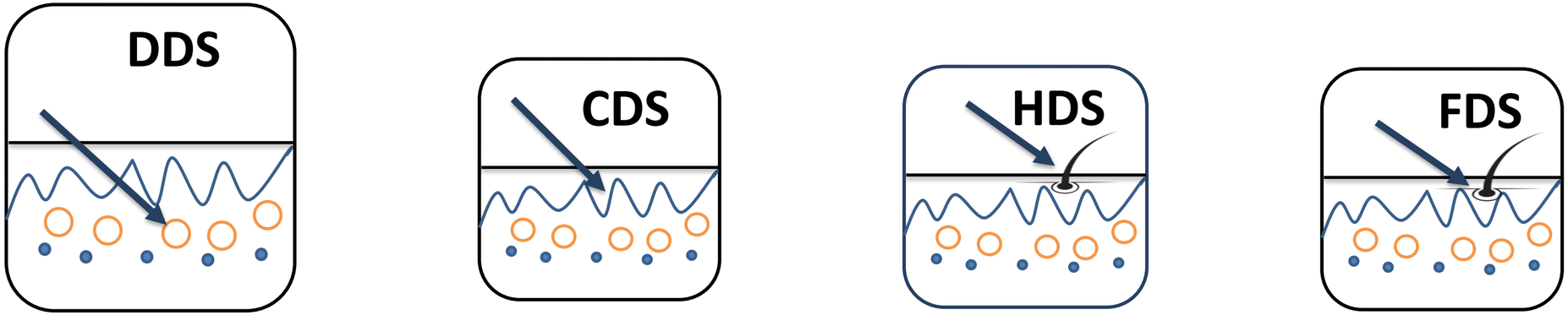


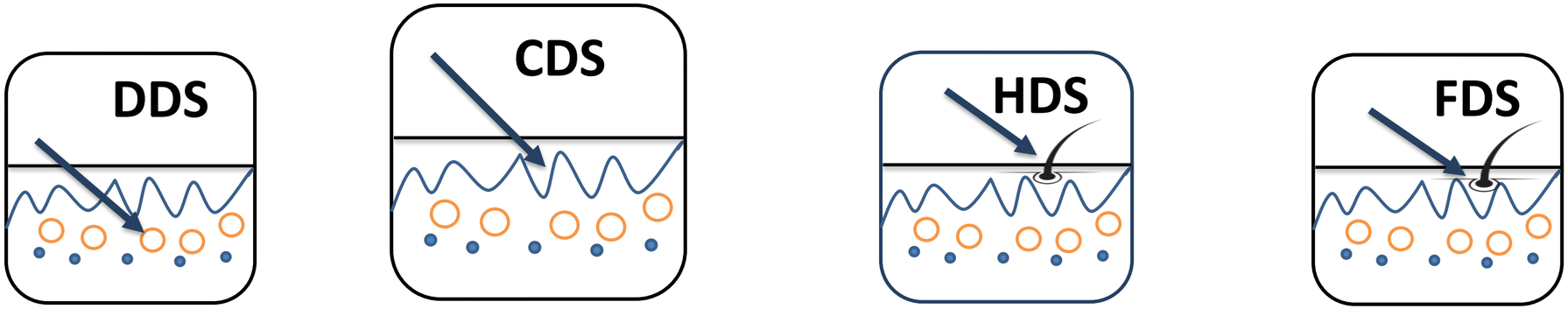

DDS
CDS
HDS
FDS

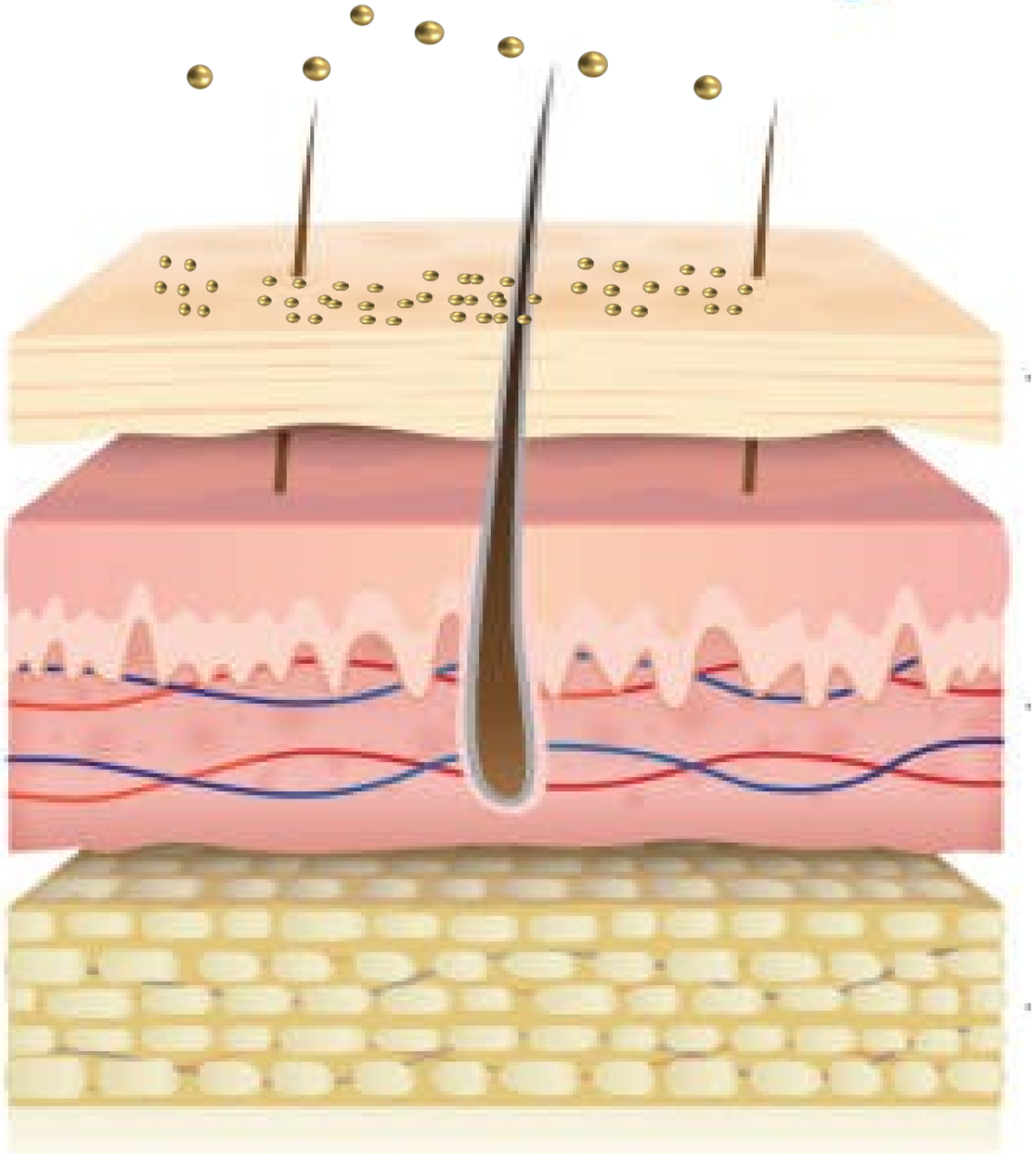


DDS-DEEP DELIVERY SYSTEM

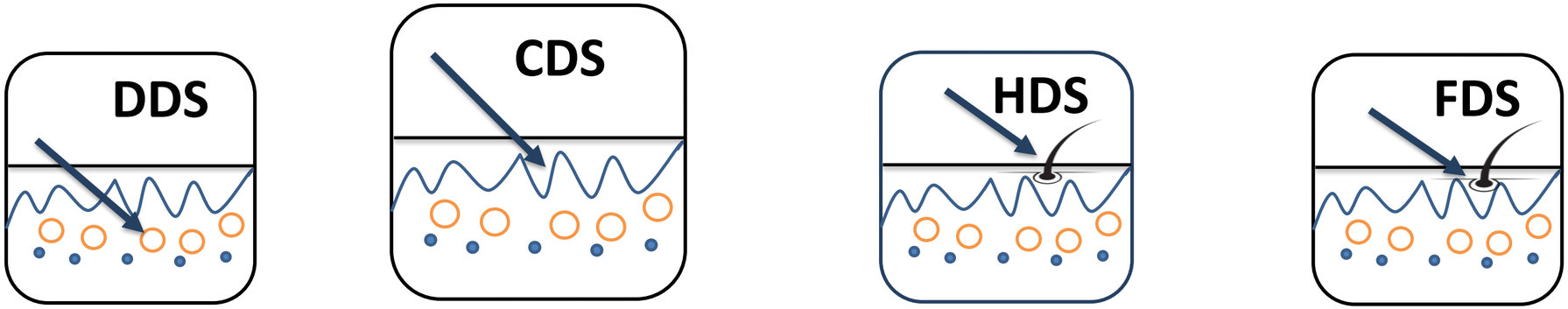
is capable of penetrating the innermost skin layers and selectively delivering the active ingredient

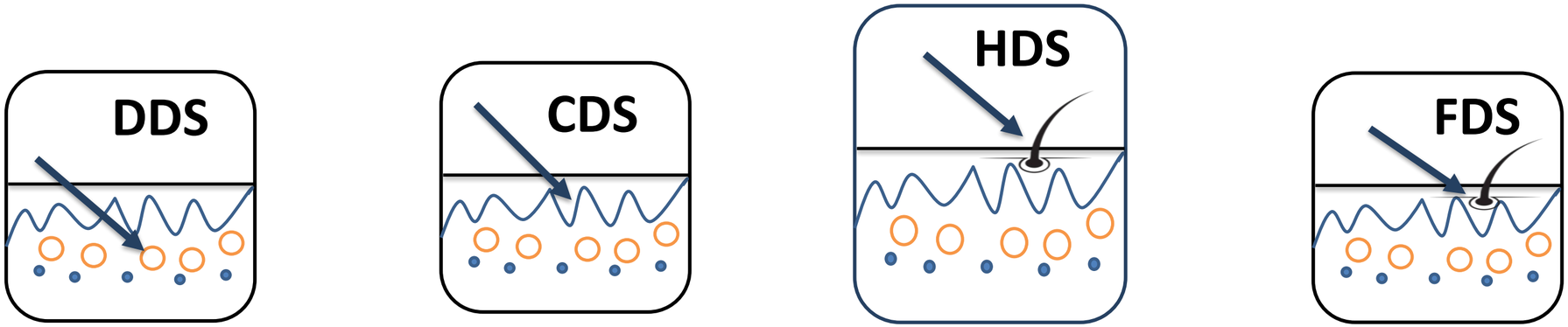

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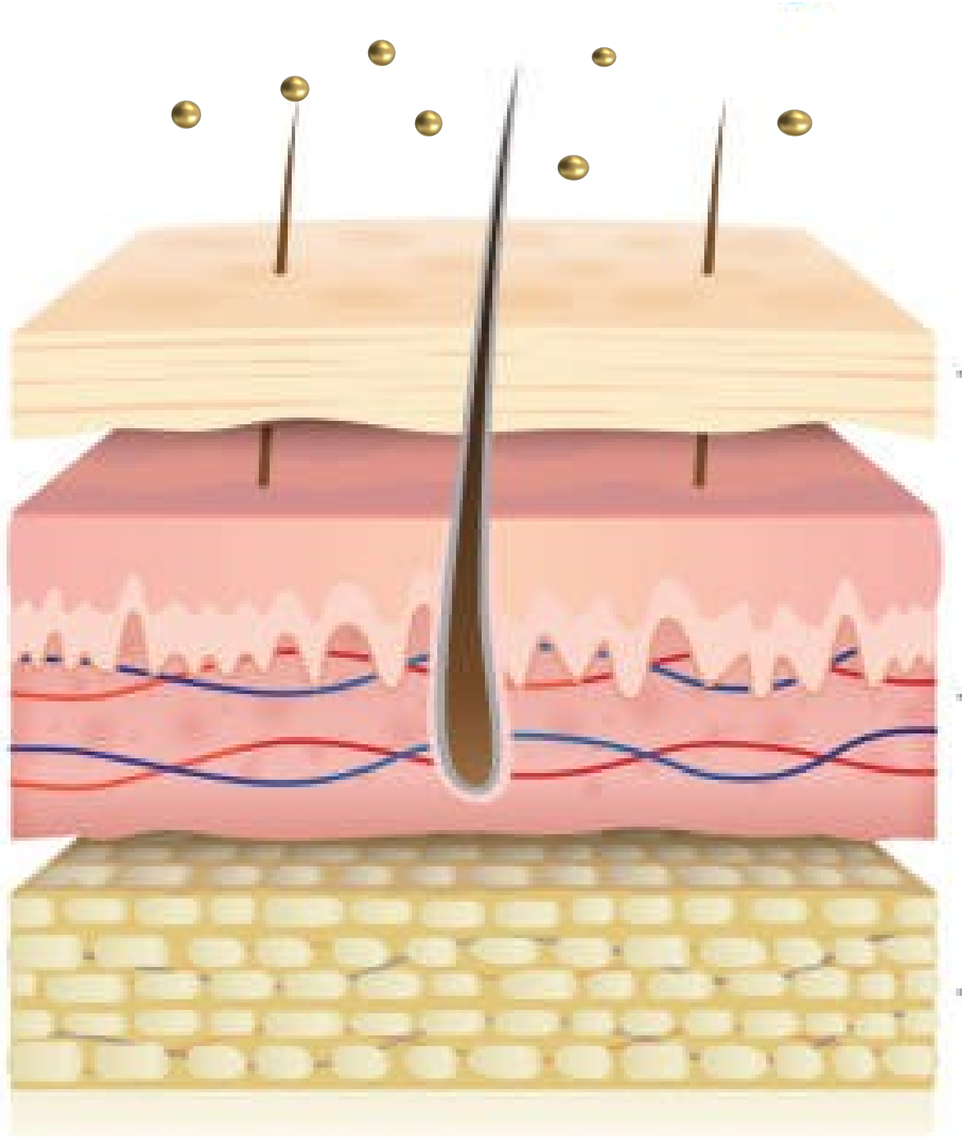

DDS
CDS
HDS
FDS



CDS- CORNEUM DELIVERY SYSTEM
Minimizing the penetration of the active ingredient and concentrating it on the outermost skin layers

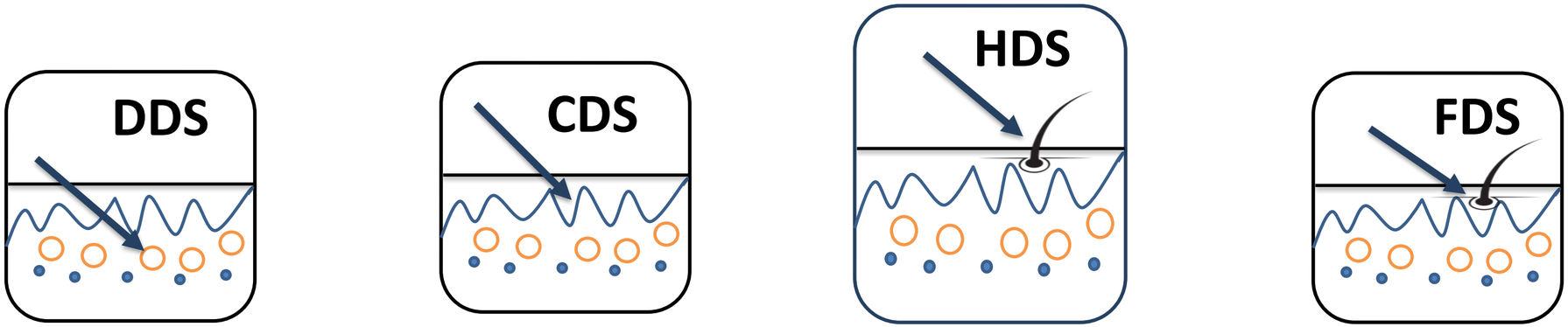

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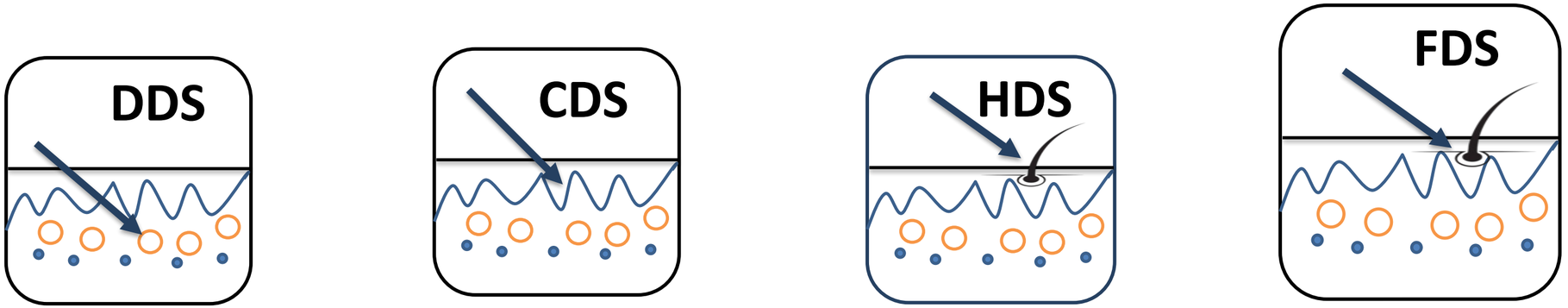

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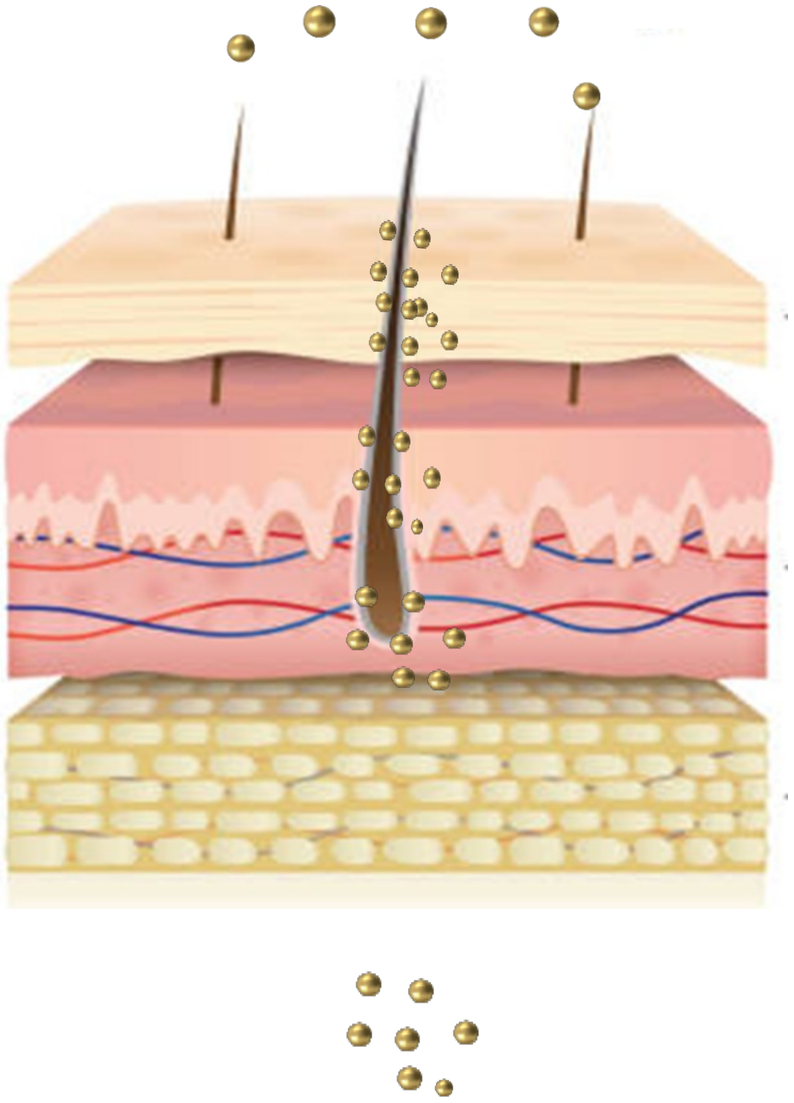


HDS-HAIR DELIVERY SYSTEM

NP adhere themselves to the hair stem and progressively deliver the active ingredients they contain to the hair cuticle for its part


DDS
CDS
HDS
FDS


DDS
CDS
HDS
FDS



FDS- FOLLICULAR DELIVERY SYSTEM

selectively Vehicles the active ingredients to the hair bulb and the sebaceous glands

Can you guess what I'm going to talk about next?



Why we need sunscreens?

1. The skin's natural defense mechanisms, such as melanin production, which prevents the penetration of UV radiation into the deeper skin layers, decreasing the release of ROS and skin photosensitization, are insufficient for an effective skin protection. In addition, melanin indirectly scavenges reactive oxygenated species (ROS) formed during the UV-inducing oxidative stress on the skin.
2. The amounts of melanin in the skin depend on the phenotype. In most phenotypes, endogenous melanin is not enough for full protection, especially in the summertime.
3. Therefore, the use of topical sunscreens and additional forms of protection is essential to prevent health issues.

Sunscreens



1. UV filters can be chemical or physical
2. A suitable alternative to chemical filters is the use of physical UV filters (zinc oxide [ZnO] and titanium dioxide [TiO2]). But, they result in white cast on skin.
3. A way to overcome this problem is to decrease its particle size and/or incorporate these filters in nanoparticles
4. But titanium dioxide and zinc oxide nanoparticles were shown to trigger the formation of free radicals upon exposure to UV rays
5. To overcome this, nanoparticles were coated with materials that almost completely block the release of free radicals.

SHEER AND LIGHTWEIGHT
HYALURONIC ACID
TRANSPARENT ZINC OXIDE
UVA/UVB PROTECTION







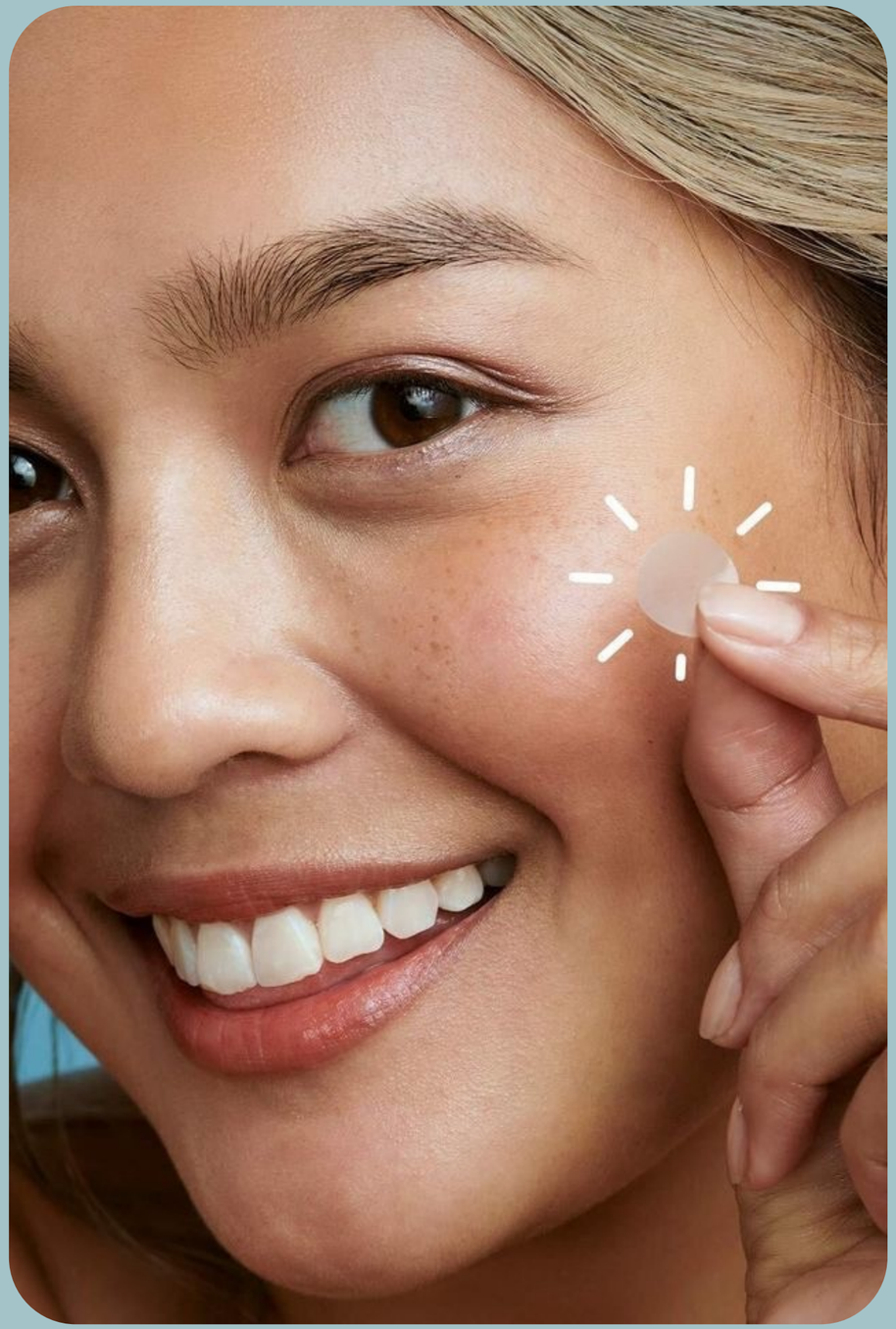
ACNE

1. Acne vulgaris is one of the most commonly observed skin diseases, affecting young adults (11–30 years). It can cause permanent physical scarring.

2. Often associated with acne-related bacteria such as Propionibacterium acnes, Staphylococcus epidermidis, Staphylococcus aureus

3. There are two types of acne that you can get, inflammatory and non-inflammatory acne
   - Non-inflammatory: on the surface of the skin
   - Inflammatory: is from the deeper layers of the skin







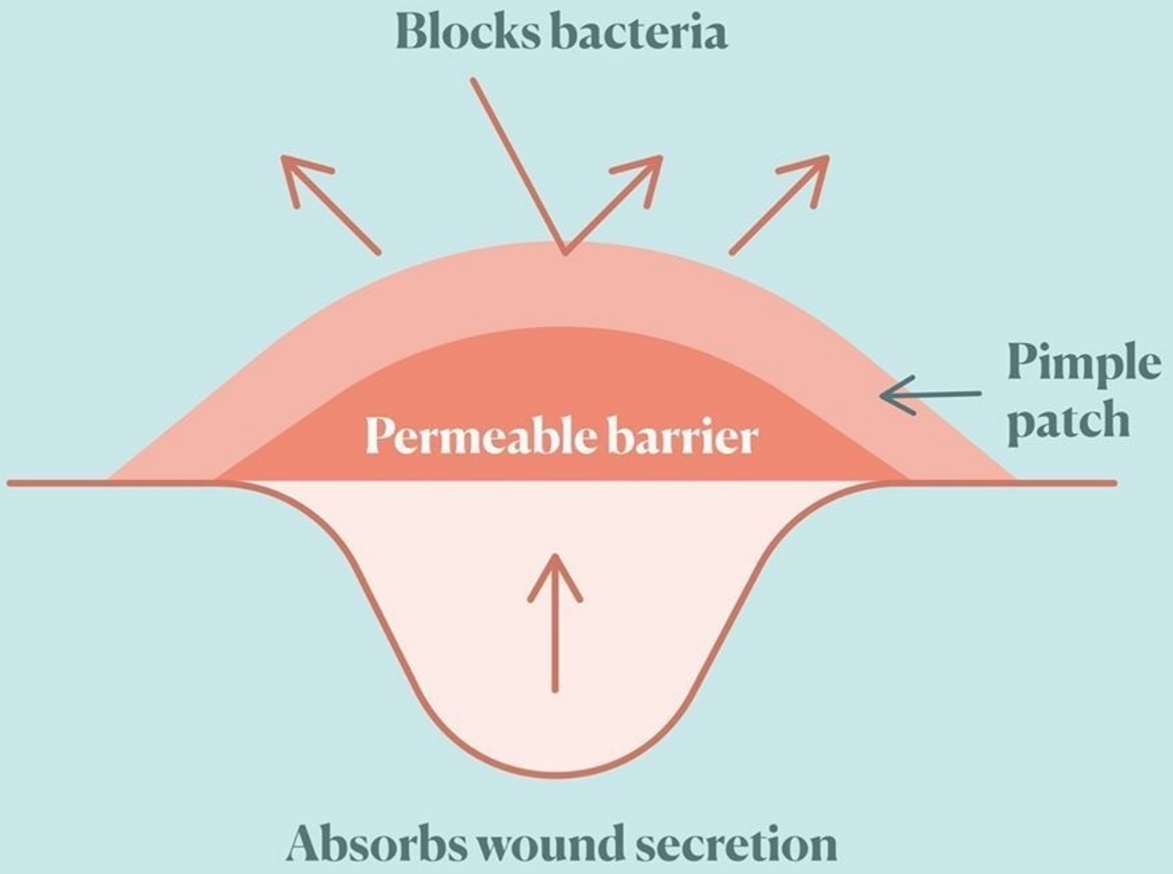


PIMPLE PATCHES

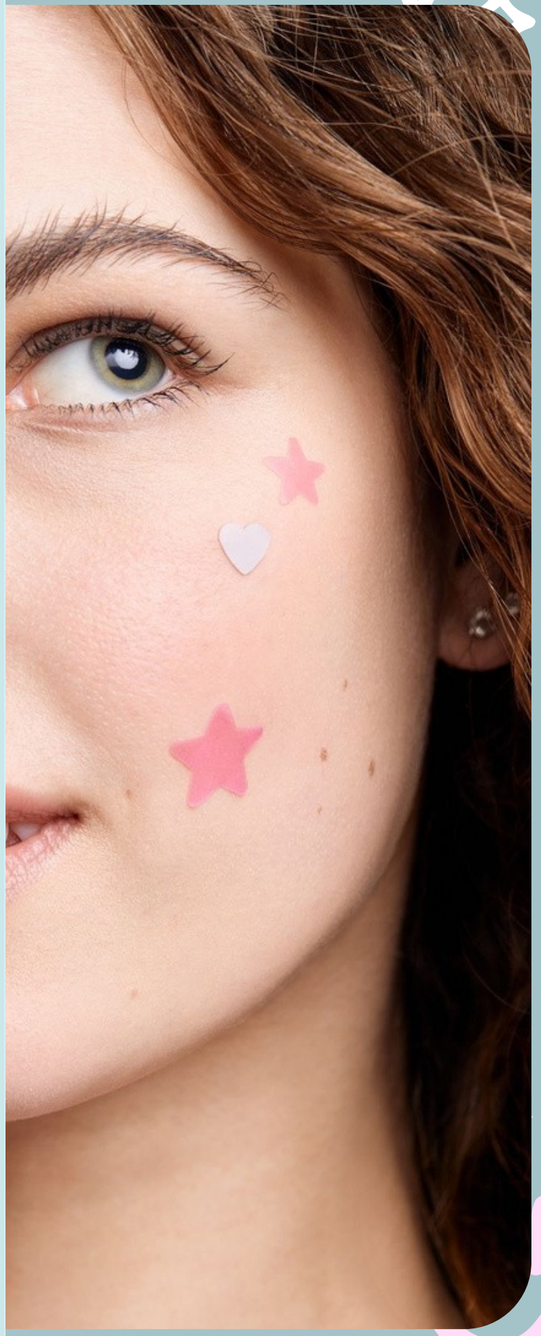




How Hydrocolloid Works



@theklog



ACNE PATCH BASE ON NANOMATERIAL GELATIN/CHITOSAN BILAYER WITH CENTELLA ASIATICA

Making gelatin and chitosan NP solution

Lyophilization of Centella asiatica (CA) to obtain dry plant extract

Cytotoxicity Assay of CA on Human skin fibroblasts

Screening of CA for antibacterial activity by disk diffusion analysis

Preparation of Gelatin/Chitosan Solution Loaded with PA/CA Extracts

Making of Mono and Bilayer Patches

Morphological Characterization of the Patches

Bilayer GC patch's composition analysis by FTIR

Water Retention Assay of patches

Antibacterial Activity Assay of patches

Conduction of skin irritation test on rabbits


BREYLEE®
ACNE PIMPLE PATCH
Acne Healing & Blemish Spot Treatment
Made with Tea Tree Exact
10mm
12 Count
12mm
10 Count
0.3mm
Perfect for fast healing
Hydrophilic Colloid
Transparent dot
Effective antibacterial protection
22
Patches

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thank
you